COMPILATION

```
@Seodas ➜/workspaces/NASM (main) $ nasm -f elf32 calculator.asm -o calculator.o
@Seodas ➜/workspaces/NASM (main) $ gcc -m32 -no-pie calculator.o -o calculator
/usr/bin/ld: warning: calculator.o: missing .note.GNU-stack section implies executable stack
/usr/bin/ld: NOTE: This behaviour is deprecated and will be removed in a future version of the linker
@Seodas ➜/workspaces/NASM (main) $ ./calculator

------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice:
```

TEST CASE ADDITION

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: A
Enter first number: 12
Enter second number: 15
The sum is 27
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: A
Enter first number: 623
Enter second number: 3968
The sum is 4591
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: a
Enter first number: 389
Enter second number: 9068340
The sum is 9068729
```

TEST CASE SUBTRACTION

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: S
Enter first number: 736
Enter second number: 53
The difference is 683
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: s
Enter first number: 65
Enter second number: 6464
The difference is -6399
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: s
Enter first number: 37865
Enter second number: 7832
The difference is 30033
```

TEST CASE MULTIPLICATION

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: m
Enter first number: 54
Enter second number: 5
The product is 270
```

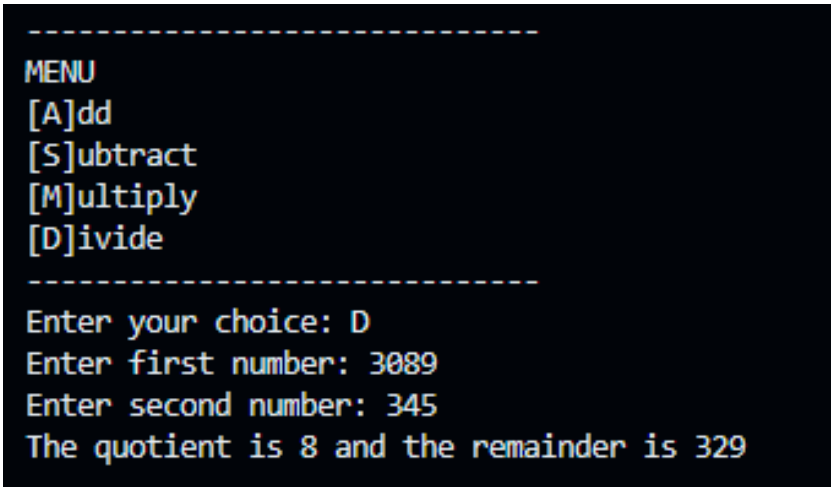
```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: m
Enter first number: 342
Enter second number: 3
The product is 1026
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: m
Enter first number: 6326
Enter second number: 568
The product is 3593168
```

TEST CASE DIVISION

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: D
Enter first number: 5223
Enter second number: 23
The quotient is 227 and the remainder is 2
```

```
------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
------------------------------
Enter your choice: d
Enter first number: 53867
Enter second number: 9768
The quotient is 5 and the remainder is 5027
```



EXPLANATION

The program displays a menu and reads the user's choice. The choice determines which arithmetic procedure runs. The read_numbers procedure asks for two integers, and the selected procedure calculates and displays the result. The division procedure checks whether the second number is zero before dividing. After each operation, the program asks whether to try again; entering Y clears the terminal and returns to the menu, while another response ends the program.
The %d input format reads whole numbers such as 12 and 99, so the program accepts two-digit inputs. Division uses integer arithmetic and displays both the quotient and remainder.